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PRACTICE MANAGEMENT

CRYSTAL PM

HOW-TO GUIDE

Set up a Crystal Meet room

Put a Crystal Meet device in a conference room and join Zoom and Google Meet calls from the room's screen, step by step.

This guide takes one conference room from an empty Raspberry Pi to a room that shows up in the Crystal Meet dashboard and joins calls from its own web page. Plan about 45 minutes per room, most of it waiting for downloads. No programming experience is needed: if you can follow a recipe and type the commands printed here, you can do this. Repeat it for each room. If the room's TV is an LG, the optional section The TV, after step 4, makes the table screen its remote and touchpad.

STEP 1
Prepare the device
Install Raspberry Pi OS, sign in, connect the TV and camera.

STEP 2
Create the room
Add the room on the dashboard's **Provisioning** page and copy its token.

STEP 3
Install Crystal Meet
One command with the room's config file.

STEP 4
First run and screens
Join a test call, pick the TV's screensaver, then point the table screen and the door sign at the device.

BEFORE YOU BEGIN

Hardware per room. Raspberry Pi 4 (4 GB) or Pi 5, the official power supply, a 32 GB microSD card, an HDMI cable to the room's TV, and a USB camera and speakerphone (for example a Logitech Brio 505 and a Jabra Speak2 55).

Two screens on the network. A touch tablet or PoE touch panel on the table to join and control meetings, and a small screen by the door to show whether the room is free. Neither connects to the Pi: each only needs a browser, power, and the office network. Give the room device a fixed address on your router (a DHCP reservation) so the screens can always find it.

The dashboard. The Crystal Meet dashboard must already be running on the office network; the guide "Set up the Crystal Meet dashboard" installs it on its own Raspberry Pi with a fixed address, for example `http://10.0.0.50`. Rooms talk to it on port `3001`.

Accounts. A dashboard sign-in with the admin role, and the Zoom or Google Meet links you want to test with.

1 Prepare the device

- **Flash the card.** On your computer, open **Raspberry Pi Imager**, choose your Pi model, then **Raspberry Pi OS (64-bit)** with desktop, and your microSD card.
- **Pre-fill the settings.** When Imager offers customisation, set the hostname to something like `crystal-meet-room-1`, the username to `pi` with a password you keep, your Wi-Fi network and your time zone, and turn on SSH. Write the card.
- **Connect and boot.** Put the card in the Pi, plug in the TV over HDMI, the camera and speakerphone over USB, then power. Wait for the desktop to appear on the TV.
- **Turn on automatic sign-in.** Open **Preferences › Raspberry Pi Configuration › System** and set **Auto Login** to on, so the room comes back by itself after a power cut.

You should now see the Raspberry Pi desktop on the room's TV, and the Pi should be on the office network.

2 Create the room in the dashboard

- **Open the dashboard** at its address, for example `http://10.0.0.50` or `http://crystal-meet.local`, and sign in.
- **Go to Provisioning** and fill in the **Room Name** exactly as people call the room (for example `Room 1`) and, if you like, its **Location**. Press **Generate Token**.
- **Copy the token** shown under **Enrollment Token**. It is a long string of letters and numbers; you will paste it into the device's config in the next step.

You should now see the token on the page and, on **Devices**, the room listed as provisioning until its device connects.

> **GOOD TO KNOW**
>
> The token links one device to this room and works once. If a device is ever reinstalled, create a new token for it. Nothing else about the room needs to change.

3 Install Crystal Meet on the device

- **Open a terminal** on the Pi (the black icon in the top bar), or connect over SSH from your computer.
- **Get the software.** Type `sudo apt update && sudo apt install -y git`, then `git clone https://github.com/ben-abeo/croom.to.git`, then `cd croom.to`.
- **Make the room's config.** Type `cp deploy/rooms/room-1.yaml ~/room.yaml` (use `room-2.yaml` or `room-3.yaml` for the other rooms), then `nano ~/room.yaml`.
- **Fill in three values.** The room's **name** and, if people call the place something, its **location** (delete that line to show the name alone); the dashboard **url** as `http://<dashboard address>:3001`; and the **enrollment_token** you copied. Save with **Ctrl+O**, Enter, then **Ctrl+X**.
- **Run the installer.** Type `sudo bash installer/install.sh --config ~/room.yaml`. It installs the browser and the room software; expect 10 to 20 minutes.
- **Start it.** When the installer prints **Crystal Meet installation complete**, type `sudo systemctl start croom`. The service also starts on every boot from now on. If it ever stops, the **Start Crystal Meet** icon on the device's desktop (over VNC) starts it again.

  You should now see the room's screensaver fill the TV: the room's name, its status and the time. The room's TV needs no keyboard or mouse, and nobody has to touch it.

4 First run and the screens

- **Open the room page.** On your laptop or phone on the office network, go to `http://crystal-meet-room-1.local:8080`, or use the device's IP address with `:8080`.
- **Check the dashboard.** On **Devices**, the room should show **online** within a minute.
- **Join a test call.** Paste a Zoom link, or a Zoom meeting ID, into **Join with a link** and press **Join**. A meeting ID, or a Zoom link without its passcode, makes a **Passcode** field appear under the box: type the meeting's passcode there before you press **Join** (on the table screen, tapping the field opens the on-screen keyboard on its digits). The TV shows the meeting joining; the page switches to the meeting screen, which offers **Mute**, **Turn camera off** and **Leave**.
- **Leave.** Press **Leave**, then **Tap again to leave**. The TV returns to the screensaver.
- **Pick the TV's idle look.** Tap **Options**, then **Screensaver** on the room page, and choose **Information** (name, status, next booking and the time), **Quiet** (name and status only), **Brand** (the logo) or **Bounce** (the logo drifting around the screen). The TV changes at once and the choice is remembered.
- **Connect the calendar.** To show real bookings and let people join them, follow the guide **Connect Crystal Meet rooms to Google Calendar** next. Until then the room page works with pasted links only.
- **Connect Zoom.** To join Zoom meetings, including ones hosted by other companies, follow the guide **Connect Crystal Meet rooms to Zoom**. Without it, Zoom turns the room away as an automated guest.

- **Set up the table screen.** On a Raspberry Pi with the touch display, clone the repository as in step 3 and type `sudo bash installer/install-kiosk.sh --url http://<device address>:8080/`. The room page opens full screen at every login with its own on-screen keyboard, and a **Room controls** icon on the desktop reopens it if it is ever closed; run the command again to change the address. After updating Crystal Meet, pull the repository on the table Pi too, run the install-kiosk command again and reboot it so the new scale takes effect. On a tablet, open its kiosk browser (Fully Kiosk Browser on Android, Guided Access with Safari on an iPad), set the start page to `http://<device address>:8080/`, turn on full screen and keep the screen awake.
- **Type on the table screen.** Tap the link box (or the passcode field) and the on-screen keyboard comes up from the bottom of the screen. It has three layers, like an iPad's: letters first, then **123** for numbers and the symbols a link or a passcode needs, then **#+=** for the rest; **ABC** goes back to the letters. Tap the shift key (the up arrow) once for one capital, or twice quickly to lock capitals on, as caps lock does, and a dot shows under its arrow; tap it again to let go. Every key answers the moment a finger lands, so both thumbs can type at once, and holding the delete key keeps deleting. Each key clicks softly, and a letter or symbol shows a larger copy of itself above the key while you press it. **Join** on the keyboard joins the meeting you typed; the key with the down arrow puts the keyboard away, and so does a tap anywhere else on the page. To turn the click off, tap **Options**, then **Key click off** under **Keyboard**; **Key click on** brings it back. The room remembers the choice, for every screen that opens its page.
- **Set up the door sign.** On the door screen, set the start page to `http://<device address>:8080/sign`. It shows green when the room is free, amber when a meeting is about to start, and red while the room is in use or booked.
- **Set up the camera and sound.** Tap **Options**, then **Camera framing** (the TV shows the camera between meetings; a screen left alone returns to the idle screen after ten minutes, and **Back** returns at once), press **Find the stops**, nudge with the arrows and zoom, then **Save as Home**. Frame the table, press **Save**, then a preset; repeat for the whiteboard. Tap a preset to return to it, even in a call. Moves take seconds (**Moving…**); the camera goes Home at every service start. During a call the page shows the meeting screen on its own: **Mute**, **Turn camera off**, **Leave**, the volume, **Camera** for a preset or a nudge, and **Upcoming meetings** to see what is booked next. On that screen, under **Sound**, set the level and try **Mute speaker**; `audio.output_device: HDMI` in the room config (then restart) sends the sound to the TV. **No speaker found** or **No controllable camera found**: the device sees no speaker, or no camera that pans or zooms (the MeetUp does).

  You should now see the room page say the room is free again, the door sign green, and the dashboard still showing the device online.

> **TIP**
>
> Zoom links that include their passcode (the part after `?pwd=`) join straight through. Google Meet needs the room signed in once as its own Workspace user; the guide "Let Crystal Meet rooms join Google Meet" covers it.

OPTIONAL, FOR AN LG TV

The TV

If the room's TV is an LG webOS TV (the Boardroom's LG C6, for example), the table screen can be its remote and a touchpad, in a meeting as well as outside one, so nobody needs the TV's own remote or a mouse in the room. The room device and the TV must be on the same network. You set the TV up once per room. A room with no LG TV still has the touchpad for the room device's mouse: under **Options**, then **TV**, the card that says **No TV is set up for this room** keeps the **Full-screen touchpad** button, and the touchpad's strip says the same.

Set it up

- **On the TV, once.** With the TV's own remote, open **Settings › General › Devices** and turn on **Mobile TV On** (some TVs call it Turn on via Wi-Fi) and **LG Connect Apps**. On webOS 25 they are under **TV Management** or **External Devices**; menus move between versions, so if yours differs, look through the TV's settings for those two names. Turn on **Quick Start+** as well if the TV has it: the TV then keeps its network connection while it is off, and the device can tell it to turn on over that connection, so Mobile TV On may stay off. Without Quick Start+ a TV that is fully off can only be woken by Wake-on-LAN, a wake-up message sent over the network, which needs Mobile TV On. The device always sends that message first, so the TV's MAC address must be in the config either way.
- **Find its address.** On the table screen tap **Options**, then **TV**, then **Find TVs**. It lists the LG TVs that answer, each with the name the TV announces on the network and its address. The MAC address is not listed: read it in the TV's network settings or on your router's list of connected devices (a TV that has a cable and Wi-Fi has a MAC address for each, so use the one for the connection it is on). Give the TV a fixed address on your router, as for the room device, so that the address you write down stays right. If you already know the address, skip Find TVs.
- **Tell the device about the TV.** On the device type `sudo nano /etc/croom/config.yaml` and change the **tv:** block that is there (the room files ship it with **enabled: false**) to read as below; add the block at the end only if the file has none: with two, the last one silently wins and the other is ignored. Replace `10.0.0.41` and `AA:BB:CC:DD:EE:FF` with your TV's address and MAC address: the example MAC address is a valid one, so a block pasted as it is warns nowhere and wakes nothing. Save with **Ctrl+O**, Enter, then **Ctrl+X**, and type `sudo systemctl restart croom`.

```
tv:
  enabled: true
  host: "10.0.0.41"
  mac: "AA:BB:CC:DD:EE:FF"
  pi_input: "HDMI_1"
  keep_on: true
```

- **What the lines mean.** host is the TV's address. mac is its MAC address, which every Turn on needs, with Quick Start+ too; keep the quotes, because YAML reads a MAC address of digits only, such as `001122334455`, as a number. pi_input is the TV input the room device's HDMI cable is plugged into, `HDMI_1` for the first socket and `HDMI_2` for the second (the check below lists the TV's inputs). keep_on turns the TV on when the service starts, before every join and about 5 seconds after anything but the table turns it off (its own remote, for example); before every join it also switches the TV to pi_input, even from live TV or an app, and leaves a TV alone whose state it cannot read. A join waits up to 15 seconds for the TV to wake, with the page showing Joining, and then goes ahead without it. An Off from the table holds until someone presses On or the next join begins, and restarting the service forgets it.
- **Pair from the table.** Tap Options, then TV. The screen says the room is set up for the TV at its address. Tap Pair, then accept the request that appears on the TV, with the TV's own remote, within a minute. You do this once: the device keeps the key the TV hands out in its data folder, normally `/var/lib/croom/tv-client-key`, readable only by the user the service runs as, and the remote opens.
- **Check it from the device.** Type `/opt/croom/venv/bin/croom --check-tv -c /etc/croom/config.yaml` as the user the service runs as, the one who ran the installer (from another account, put `sudo -u pi` in front, with that user's name), and never as root: the TV's key file is private to that user (mode 600), and a check run as root would leave a renewed key where the service cannot read it. It reads:

  ```
  TV: 10.0.0.41, MAC AA:BB:CC:DD:EE:FF
  The TV is paired with this device (its key is in /var/lib/croom/tv-client-key).
  The TV answers: LG C6, on.
  Inputs: HDMI 1 (HDMI_1), HDMI 2 (HDMI_2).
  The TV shows HDMI 1 (HDMI_1), the room Pi's input (tv.pi_input).
  Ready: the TV answers.
  ```

- **What else it says.** The check only looks: it does not wake or pair the TV, and leaves nothing connected. A line beginning Warning (a MAC address that is missing or is not one, a pi_input the TV does not have, a TV showing another input) means the TV answers but something will not work as set, and the last line then reads "The TV answers, with the warnings above." When the TV does not answer, the check says why: it is not paired for this user, it does not answer, or it no longer accepts the device. If the TV shows its pairing prompt during the check and someone accepts it, the TV hands out a new key; the check keeps it and says "The TV issued a new key (its pairing prompt was accepted during the check), which is kept." It exits with 0 when the TV answered and 1 when it did not. With no TV in the config it lists the LG TVs it finds on the network instead, as Find TVs does.

  You should now see the check end with "Ready: the TV answers." and the table's remote open on the TV's name.

Using it

- **Open the remote.** Tap Options, then TV. During a meeting tap TV on the meeting screen, beside Camera and Upcoming meetings. Back returns to the screen you came from, and the remote returns by itself after ten minutes with no touch, or two minutes during a meeting.

- **What is on it.** The top line names the TV and says what it shows, for example **Boardroom TV · On · HDMI 1 (Crystal Meet)**. The name is the one the TV gives over its control connection, often its model name such as LG C6, so it can differ from the name Find TVs listed for the same TV. Under it are **Power**, **Input**, **Home** and **Menu**; the volume with **Mute**; the channel keys; a round d-pad with **OK** and, below it, **Back**, **Exit**, **Info** and **Guide**; the number keys with **Go**; the playback keys and the four colour keys.
- **Power, Input and the second taps.** **Power** asks for a second tap to turn the TV off, as Leave does. When the TV is off, the top line shows **Turn on** in place of its state, and turning it on takes up to about 23 seconds (the line says Turning on…); the other keys wait for it. **Input** opens a row of the TV's inputs, with the room device's marked **Crystal Meet**; a pick switches the TV at once. A key the TV refuses flashes amber once and the reason appears under the top line. **Forget TV**, at the right end of the top line, asks for a second tap too.
- **The touchpad modes.** Three buttons sit under the keys. **Pointer** turns the d-pad into a trackpad for the TV's own cursor: slide one finger to move it, tap to click, slide two fingers to scroll, and press and hold, then move, to drag. Tap **Pointer** again to bring the d-pad back. **Room Pi mouse** does the same for the Pi behind the TV: the Pi's desktop takes its virtual mouse like a USB mouse, so its pointer should appear on the TV (the room's first use confirms it), and you can click a button on whatever the Pi is showing, such as a dialog in a meeting. A two-finger tap there is a right-click.
- **The pad's border and its link.** A blue border on the pad (on the full-screen touchpad, a thin blue line round the screen's edge) means a hold is armed: after a short rest the next move drags, a touch that moves sooner is a move for good, and after a rest a nudge under 8 pixels does nothing. If the link to the device drops, the pad (the full-screen strip too) says **Reconnecting…** and the page tries again after 1 second, then 2, then every 5 seconds; a finger moves nothing until it is back.
- **The full-screen touchpad.** **Full-screen touchpad** gives the whole table screen to the pointer, with the same gestures. A strip names what it moves, with a switch between **TV** and **Room Pi** and a **Done** button; nothing else is on the screen to be tapped by mistake. It returns to the remote when you tap **Done**, or by itself after ten minutes with no touch. How the gestures feel, how far the cursor runs for a finger's travel and how long a tap may last, is tuned on the table's own screen.

FOR THE OPERATOR

A pairing waits 60 seconds for the person at the TV and gives the TV 5 seconds more to finish; Cancel, or Forget TV, ends it at once. While a TV is waking, which a join gives up to 15 seconds, the device sends the wake-up message again every 3 seconds. Restarting the service ends a pairing, a Turn on and an open touchpad within seconds, because it no longer waits for the TV. A TV that is not answering is reported in words under the remote's top line, for example "The TV is not answering at 10.0.0.41". Unpair with Forget TV, not by deleting the key file: a pointer that is already connected keeps working until its link drops. The touchpad's events travel over a WebSocket at `/ws/pointer` on port 8080, which like the rest of the room page takes no sign-in.

FOR EVERY ROOM DEVICE

Updating a room device

Update a device that already runs Crystal Meet in the order below. Every room device needs these steps, with an LG TV or not, because the library behind the touchpad's mouse is a hard dependency and is compiled on the device. The first and third steps are needed once and do no harm when repeated.

- **Install the build tools.** Type `sudo apt-get install -y python3-dev gcc`. Without them the update stops with an error while it builds the mouse library.
- **Update the software.** Type `sudo /opt/croom/venv/bin/pip install -q --upgrade git+https://github.com/ben-abeo/croom.to.git`.
- **Set the virtual mouse up.** Type `cd ~/croom.to && git pull`, then `sudo bash -c 'source installer/install.sh && install_virtual_mouse'`. That is the installer's virtual-mouse step on its own: a udev rule hands /dev/uinput to the input group, the uinput module is set to load at every boot, and your user joins the input group. (Running the whole installer again includes it.)
- **Restart.** Type `sudo systemctl restart croom`: a new group only reaches a service that starts afresh.

IF SOMETHING IS OFF

Troubleshooting

The room page does not load

On the Pi, type `sudo systemctl status croom`. If it is not running, type `sudo journalctl -u croom -n 50` and look for the first line marked ERROR. Make sure port 8080 is not used by something else on the device.

The device shows offline

From the Pi, type `curl http://<dashboard address>:3001/health`. If nothing comes back, the dashboard is not reachable from the room's network: check the address in `~/room.yaml` and that the dashboard Pi is up (its guide has the checks). If the token was already used, create a new one and run the installer again with the updated config.

The browser opens but never joins

Zoom needs the meeting's passcode: paste the full invitation link, or type the passcode in the **Passcode** field that appears for a meeting ID or a link without one (on a "Couldn't join" card that says "Zoom needs the meeting passcode", press **Join again**). Google Meet needs the room signed in (the guide "Let Crystal Meet rooms join Google Meet"); the room page says so when the sign-in is missing or expired. The room page quotes what Meet showed and the log names a screenshot under `/tmp`; from the Pi, `PLAYWRIGHT_BROWSERS_PATH=/opt/croom/browsers DISPLAY=:0 /opt/croom/venv/bin/croom --check-meet <link>` reports the same (stop the service first: `sudo systemctl stop croom`).

No sound or picture in the call

Check the USB camera and speakerphone are connected before the device starts, then reboot the Pi. The service runs as the signed-in desktop user so it can use the room's screen and audio. From the Pi, `PLAYWRIGHT_BROWSERS_PATH=/opt/croom/browsers DISPLAY=:0 /opt/croom/venv/bin/croom --check-meet <link>` (stop the service first: `sudo systemctl stop croom`) prints "Devices the browser sees" with the microphones, speakers and cameras the browser can use; if the MeetUp is missing from "Devices the browser sees" but `wpctl status` lists it, the browser cannot reach PipeWire: make sure `libpulse0` is installed (the installer adds it) and that the service runs in the desktop user's session.

No browser window opens on the TV

The service waits for the desktop before it starts, so give a freshly booted Pi a minute. If the screensaver still does not appear, type `sudo journalctl -u croom -n 50` and look for "Failed to initialize". The browser is installed under `/opt/croom/browsers`; if that folder is missing, run the installer again.

The screens cannot open the page

Use the device's IP address rather than its `.local` name; many tablets do not resolve those. Check that the screen and the device are on the same network, then open `http://<device address>:8080/api/status` from a laptop to confirm the device answers.

The TV is not found

Find TVs lists nothing, or the table says "The TV is not answering at 10.0.0.41". Each TV answers the device's search straight back to the device, so a firewall on the device that blocks incoming traffic by default hides every TV: allow incoming UDP from the TV's address, or skip Find TVs and read the address in the TV's network settings. The search and the wake-up messages leave by the device's default network connection, so the device and the TV must be on that same network. Check that the TV is on, that **LG Connect Apps** is on and that **host** is the TV's current address, then type `/opt/croom/venv/bin/croom --check-tv -c /etc/croom/config.yaml` to see what the device sees.

The TV will not wake

Turn on says the TV did not turn on, or a join goes ahead with the TV dark. A MAC address in **mac** that is missing or is not one stops every wake, Quick Start+ or not, and the check warns about it. A TV that is fully off, with Quick Start+ off, can only be woken by Wake-on-LAN: **Mobile TV On** must be on and the MAC address must be the TV's own (a TV that has a cable and Wi-Fi has one for each). With **Quick Start+** on, a TV that answers is told to turn on over its connection, so Mobile TV On and a wrong but well-formed MAC address do not matter. A join waits up to 15 seconds for the TV, with the page showing Joining, and then goes ahead without it.

The TV no longer accepts the table

The table says "The TV no longer accepts this table", or the TV keeps showing a request to allow a device: it has forgotten this table, after a reset or when its list of paired devices was cleared. Tap **Forget TV** (on the remote it asks for a second tap), then **Pair**, and accept the new request on the TV with its own remote. While the TV shows a request the device waits for an answer; it stops waiting when the request is answered, when the TV goes away, or on Forget TV.

The room Pi's mouse does nothing

Pointer moves the TV's cursor but **Room Pi mouse** moves nothing, and the table says the mouse "is not allowed to use /dev/uinput" or "needs /dev/uinput, and it is not there". The device makes its virtual mouse on `/dev/uinput`, which needs `python3-dev` and `gcc` on the device before the software is installed (pip builds the mouse library), this udev rule, `KERNEL=="uinput", GROUP="input", MODE="0660"`, in `/etc/udev/rules.d/99-croom-uinput.rules`, the service's user in the `input` group, and the `uinput` module loaded; the installer's virtual-mouse step sets up the last three. Check with `ls -l /dev/uinput` (group input), `groups pi` (your user name instead of pi) and `lsmod | grep uinput`. Run the virtual-mouse step from the update recipe above, then `sudo systemctl restart croom`. The mouse is made when the first touchpad mode is turned on after the service starts; its pointer should then appear on the TV.

GOOD TO KNOW

Under the hood the device software is still called `croom`: that is the name you will see in service and log commands. Everything people see says Crystal Meet.